sound particularly exotic, negative pressure can result in something extraordinary from the point of view of general relativity: *whereas positive pressure contributes to ordinary attractive gravity, negative pressure contributes to "negative" gravity, that* is, to repulsive gravity![6]

With this stunning realization, Einstein's general relativity exposed a loophole in the more than two-hundred-year-old belief that gravity is always an attractive force. Planets, stars, and galaxies, as Newton correctly showed, certainly do exert an attractive gravitational pull. But when pressure becomes important (for ordinary matter under everyday conditions, the gravitational contribution of pressure is negligible) and, in particular, when pressure is negative (for ordinary matter like protons and electrons, pressure is positive, which is why the cosmological constant can't be composed of anything familiar) there is a contribution to gravity that would have shocked Newton. *It's repulsive.*

This result is central to much of what follows and is easily misunderstood, so let me emphasize one essential point. Gravity and pressure are two related but separate characters in this story. Pressures, or more precisely, pressure differences, can exert their own, nongravitational forces. When you dive underwater, your eardrums can sense the pressure difference between the water pushing on them from the outside and the air pushing on them from the inside. That's all true. But the point we're now making about pressure and gravity is completely different. According to general relativity, pressure can indirectly exert another force—it can exert a gravitational force—because pressure contributes to the gravitational field. Pressure, like mass and energy, is a source of gravity. And remarkably, if the pressure in a region is negative, it contributes a gravitational *push* to the gravitational field permeating the region, not a gravitational pull.

This means that when pressure is negative, there is competition between ordinary attractive gravity, arising from ordinary mass and energy, and exotic repulsive gravity, arising from the negative pressure. If the negative pressure in a region is negative enough, repulsive gravity will dominate; gravity will push things apart rather than draw them together. Here is where the cosmological constant comes into the story. The cosmological term Einstein added to the equations of general relativity would mean that space is uniformly suffused with energy but,